

G00L2a/b
Time-to-event analysis of
economic, biomedical and industrial data
2025–2026

The objective of the project is to apply the models, methods, software, ... seen in this course on a real data set of your own choice. This will allow you to get more familiar with the techniques seen in this course and to get a better understanding of the different chapters. Here are the main steps for the project:

1. Find appropriate data.
2. Analyse the data.
3. Write a report of your analysis.
4. Present this report during the exam and answer questions on the report.

Each step is explained in more detail below.

Important: You work on the project in groups of 4, 5 or 6 students. Together with your team members you analyse the data set of your group, and you write one single report. **For organizational reasons mixed groups (consisting of on-campus and blended students) are not allowed.**

1. Data acquisition

Your first task is to find appropriate data for this project. The data should satisfy the following constraints:

- The number of observations is at least 100.
- The response variable is subject to random right censoring.
- There are at least 3 explanatory variables (quantitative or qualitative), one of which is a categorical variable with 2, 3 or maximum 4 levels.
- You are free to choose data from any domain where censoring occurs (medicine, economics, actuarial sciences, sociology, astronomy,...)

Important: Every group should use a different data set. You should therefore send me an e-mail (ingrid.vankeilegom@kuleuven.be) containing as an attachment the data set that your group likes to use, together with a small description of the

context of the study, and an explanation of why you think the data are appropriate. I will then let you know whether I agree with the data you have chosen. The deadline for sending me the data set and for letting me know who are your team mates is **2 April 2026**.

Sources for finding appropriate data:

- Package ‘SurvSet’, an open-source time-to-event dataset repository: see <https://arxiv.org/pdf/2203.03094.pdf>
- Certain medical databases are freely accessible: see e.g. the SEER data base on cancer in the US (<https://seer.cancer.gov/data/access.html>) (after registration)
- Data sets available on kaggle: <https://www.kaggle.com/datasets>
- Many other websites, books, articles, etc

Preferably, don’t use data sets from R packages that are related to survival analysis, like e.g. ‘survival’, ‘KMsurv’, ‘mstate’, among others. More original and less cleaned data are more challenging and will be higher rated.

Some remarks:

- Note that a PRET evaluation is necessary in case personal data are collected and/or analyzed (see master thesis document on Toledo for more details about PRET).
- In case you like to change your data set after I have approved it (for whatever reason), you can send me another data set.
- Note that my approval does not necessarily mean that the data are 100% appropriate for the project. That is your own responsibility and judgment. I will have a rather superficial look at the data and I will make sure that the data have not been chosen by someone else.

2. Data analysis

Each data set is different and requires other kinds of analyses. The questions below are hence a starting point for your analysis, but you can add additional analyses if you think they are relevant for your data and if they make the report richer and stronger.

Part A

Before starting the analysis of the data, carry out a descriptive analysis of the variables in the data. This will allow you to get a first impression of how the data look

like, but don't forget that one of the variables is censored, which should be taken into account in the interpretation of your analysis. Transformations of the variables can take place based on these preliminary analyses.

Part B

We focus in Part B on the response variable, the censoring indicator and (one of) the categorical variable(s) (in case you have several categorical variables, take the one that is most relevant to the data).

- (a) For each of the levels of the categorical variable, compute the survival function nonparametrically. Plot them on the same graph. What do the graphs suggest?
- (b) For each level obtain an appropriate estimator and confidence interval for the 3 quartiles of the survival curves. Interpret the results.
- (c) Conduct a single test of differences between the survival curves. Justify your choice of test.

Part C

Consider now all variables in the data set.

- (a) We are interested in estimating the relative risks of survival. Assume that all relative risks are constant over time, and independent of any other covariates.
 - Build an appropriate model for these data, using the model building procedures seen in class. Make sure that the categorical variable studied in Part B is included in the model by default.
 - Find an estimator and confidence interval for the relative risks under the chosen model. Interpret these relative risks.
 - Test whether the categorical variable is significant in the obtained model. Is the result in line with your findings in Part B?
 - Is there evidence that the relative risks are indeed constant over time?
- (b) We now consider parametric regression models for the data.
 - Build an appropriate parametric model for the data, taking (for simplicity) the same variables as the ones selected for the previous model.
 - Give point and interval estimates of the regression coefficients both in the AFT model and in the linear model representation.
 - Compare the results with those under the Cox model and with those from the nonparametric analysis. Do you get the same conclusions, qualitatively speaking? Why or why not?

3. Report of your data analysis

Here are some guidelines for the report:

- Write a clear and structured report.
- Motivate well the models and procedures you use.
- Interpret the obtained results.
- Include your programmes at the end of your analysis.
- The report should not exceed 7 pages of text (tables, graphs and programmes not included).

Upload your report in Toledo (deadline is **14 May 2026** at 23:59). No reports submitted after the deadline will be accepted. When any special circumstances have occurred, you can contact the ombuds persons (Martial Luyts / Ilias Willems) *before* the deadline. The deadline in case you present the exam in August is **12 August 2026** at 23:59 (and **22 July** for the data set).

4. Presentation of your report

For the presentation, the following should be taken into account:

- Each group gives a pitch presentation of 5 minutes, followed by 10 minutes of Q&A. Make sure that you split the presentation in such a way that each team member can say something. I will tell you before the start of the presentation which student presents which part. So you should be ready to present whatever part of the presentation.
- The presentations will take place:
 - For the on-campus groups: on **21 May**, 2-4pm and 4-6pm. You listen to the presentations of the other groups in your two-hour block.
 - For the blended groups: on **22 May** in the afternoon (precise modalities to be communicated later)

5. Remarks on grading

- As explained at the start of the course, the project counts for 25% of the total score. The remaining 75% of the exam is a written open-book exam.
- You need to pass both parts of the exam. In case you fail one part but your average is 10 or more, you will receive 9/20 for your final score.

- You need to participate in both parts of the exam, otherwise you will receive NA regardless of the score of the second part.

6. Important remark regarding the use of GenAI

- Please be aware of the rules of KU Leuven regarding the use of GenAI for your project, they can be consulted [here](#).
- Be aware that if you would use GenAI for code generation, you need to be transparent about it (by clearly mentioning which parts were generated with GenAI) and you should be able to explain every line of the code during your oral presentation. So don't use blindly code generated by GenAI, but think carefully about what it does (and whether it is correct).